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On Dirichlet series satisfying Riemann's functional equation

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Summary

According to convention, Hamburger's theorem (1921) says – roughly – that Riemann's $\zeta(s)$ is uniquely determined by its functional equation. In 1944 Hecke pointed out that there are two distinct versions of Hamburger's theorem. Hecke's remark has led me, in examining just how "rough" the convention is, to prove that, with a weakening of certain auxiliary conditions, there are infinitely many linearly independent solutions of Riemann's functional equation (Theorem 1). In Theorem 1, as in Hamburger's theorem, the "weight" parameter is 1/2. In Theorem 2 we obtain stronger results when this parameter is $\geqq 2$: a "Mittag–Leffler" theorem for Dirichlet series with functional equations.

I Introduction

In this final published paper [9], Hecke describes two versions of Hamburger's theorem, on the unique determination of Riemann's $\zeta(s)$ by its functional equation. We quote from the introduction to [9] (translation mine):

"The analytic function $\varphi(s)$ of the complex variable s is determined up to a constant factor by the following conditions: Put $R(s)=\pi^{-s}\Gamma(s)\varphi(s)$.

1. With a suitable polynomial $P(s)$ suppose that $P(s)\varphi(s)$ is an entire function of finite genus.
2. Suppose $\varphi(s)$ satisfies the equation

$$R(s)=R(\tfrac{1}{2}-s).$$

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To the memory of Martin Eichler

3(a) Suppose that not only $\varphi(s)$, but also $\varphi(s/2)$, can be expanded in a Dirichlet series convergent somewhere[1]: $\varphi(s)=\sum_{n=1}^{\infty} b(n)n^{-2s}$. This condition can be replaced by

3(b) Suppose that the only pole allowed for $\varphi(s)$ is $s=1/2$; but we assume only the expressibility of $\varphi(s)$ itself as a Dirichlet series $\varphi(s)=\sum_{n=1}^{\infty} a(n)n^{-s}$, not that of $\varphi(s/2)$.

"Mr. Hamburger first proved that $\varphi(s)$ is uniquely determined by 1, 2, 3(a), and thus = const. $\zeta(2s)$" [6]; "that also 1, 2, 3(b) suffice I have proved within the framework of a general investigation, by means of reduction to the theory of certain automorphic functions [8]".

In light of Hecke's remarks, it is natural to relax both the expressibility of $\varphi(s/2)$ as a Dirichlet series in 3(a) and the restriction on the poles of $\varphi(s)$ in 3(b), and conjecture that $\varphi(s)$ is uniquely determined by 1, 2 and

3. Suppose (only) that $\varphi(s)$ can be expanded in a Dirichlet series convergent somewhere.

While this conjectured strong form of Hamburger's theorem is appealing, we demonstrate here that it fails very strongly. In fact, we prove

Theorem 1 *Let σ_0 be a real number $\geqq 1/4$. Let $\mathscr{A}(\sigma_0)$ be the space of all rational functions $A(s)$ with poles restricted to the strip $1/2-\sigma_0 \leqq \mathrm{Re}\, s \leqq \sigma_0$ and satisfying $A(1/2-s)=A(s)$. Let $\mathscr{A}_H(\sigma_0)$ be the subspace of A in $\mathscr{A}(\sigma_0)$ such that $R(s)-A(s)$ is entire for some $R(s)=\pi^{-s}\Gamma(s)\varphi(s)$ satisfying 1, 2 and 3. Then, with $n\in Z$, $n>N(\sigma_0)=[\sigma_0/2+3/8]+2$, and $A_1, \ldots, A_n \in \mathscr{A}(\sigma_0)$, some nontrivial linear combination of the A_j is in $\mathscr{A}_H(\sigma_0)$.*

Remark. Let $\mathscr{B}(\sigma_0)$ be the subspace consisting of those $A\in\mathscr{A}(\sigma_0)$ which vanish at infinity; put $\mathscr{B}_H(\sigma_0)=\mathscr{A}_H(\sigma_0)\cap\mathscr{B}(\sigma_0)$. Then Theorem 1 is, of course, still correct if we replace $\mathscr{A}(\sigma_0)$ by $\mathscr{B}(\sigma_0)$ and $\mathscr{A}_H(\sigma_0)$ by $\mathscr{B}_H(\sigma_0)$ in the statement. This formulation has the advantage that, by Hecke's version of Hamburger's theorem, distinct $A(s)\in\mathscr{B}_H(\sigma_0)$ give rise to distinct Dirichlet series $\varphi(s)$. (This does not hold for distinct $A(s)\in\mathscr{A}_H\sigma_0$).) These remarks apply equally to Theorem 2 (below).

Theorem 2 ("Mittag-Leffler") *Let k be any real number $\geqq 2$ and let $A(s)$ be any rational functions satisfying $A(k-s)=A(s)$. Then, there exists a Dirichlet*

[1] *Hecke's words are*: "$\varphi(s/2)$ soll in eine *irgendwo* konvergente Dirichlet-Reihe entwickelbar sein" (emphasis added). However, Hamburger's proof [6] requires that $\varphi(s/2)$ be absolutely convergent $\sigma>1$. Siegel [18] relaxes this condition, assuming absolute convergence for $\sigma>2-\theta$, $\theta>0$. To my knowledge, the proof of the stronger formulation of Hamburger's version, as asserted by Hecke, did not appear until 1956 [2, Theorem 7.1]

series $\varphi(s)$ convergent in some right half-plane such that $R(s)=\pi^{-s}\Gamma(s)\varphi(s)$ satisfies 1, 3,

2 − *k*. $$R(k-s)=R(s)$$

and

4. $$R(s)-A(s) \text{ is entire.}$$

Remark. (i) Theorem 2 may be termed a "Mittag-Leffler theorem" for Dirichlet series with functional equations. The proof entails a Mittag-Leffler theorem for modular forms of weight $k \geqq 2$.

(ii) The only restriction on the rational function $A(s)$, containing the pole terms of $R(s)$, is the necessary symmetry, $A(k-s)=A(s)$.

The proof of these theorems depends in an essential way upon the Riemann–Hecke correspondence between Dirichlet series with functional equations and modular forms, as extended by Bochner [1]. It relies as well upon the "generalized Poincaé series" of Eichler [4, 5, 12]. Both proofs apply the Riemann–Hecke–Bochner correspondence at the outset, to translate the question of existence of the desired Dirichlet series into a question of existence of "modular integrals" with equivalent properties. The second step is the construction, by means of the generalized Poincaré series, of the required modular integrals. Definitions are found in §III.

II The Riemann–Hecke–Bochner correspondence

The correspondence was first observed by Riemann, who used it to derive the functional equation of $\zeta(s)$ from the transformation formula of

$$\vartheta(z)=\vartheta_3(0\,|\,z)=\sum_{n=-\infty}^{\infty} e^{\pi i n^2 z}, \quad \operatorname{Im} z>0,$$

under $z \to -1/z$. However, it was Hecke – applying the Mellin transform and its inverse – who uncovered a systematic relationship between modular forms, on the one hand, and Dirichlet series with functional equations, on the other [7, 8]. In our application we need Bochner's generalization of the correspondence [1, 3]. We state this in

Lemma 3 *Let k be real, C complex and $\lambda_1, \lambda_2>0$. Suppose $F(z)$, $G(z)$ are functions holomorphic in the upper half-plane $\mathcal{H}$, defined there by the exponential series*

(2.1) $$F(z)=\sum_{n=0}^{\infty} a_n e^{2\pi i n z/\lambda_1}, \quad G(z)=\sum_{n=0}^{\infty} a_n e^{2\pi i n z/\lambda_2},$$

with a_n, b_n satisfying

(2.2) $$a_n, b_n = O(n^\gamma), \quad \gamma>0, n \to +\infty.$$

Let $\Phi(s)$, $\Psi(s)$ be the Mellin transforms of $F(iy)-a_0$, $G(iy)-b_0$, respectively:

$$\begin{cases} \Phi(s)=\int_0^\infty \{F(iy)-a_0\}\, y^s \dfrac{dy}{y}=\left(\dfrac{2\pi}{\lambda_1}\right)^{-s}\Gamma(s)\sum\limits_{n=1}^{\infty} a_n n^{-s}, \\ \Psi(s)=\int_0^\infty \{G(iy)-a_0\}\, y^s \dfrac{dy}{y}=\left(\dfrac{2\pi}{\lambda_2}\right)^{-s}\Gamma(s)\sum\limits_{n=1}^{\infty} b_n n^{-s}. \end{cases} \tag{2.3}$$

Then, assertions (A) *and* (B) *are equivalent:*

(A) For z *in* $\mathscr{H}$,

$$z^{-k}G(-1/z)=CF(z)+q(z), \tag{2.4}$$

where $q(z)$ is the "log-*polynomial sum,*"

$$q(z)=\sum_{1\leqq j\leqq J} z_j^{\alpha_j} \sum_{0\leqq t\leqq M(j)} \beta(j,t)(\log z)^t. \tag{2.5}$$

(In (2.5) $\alpha_1, \ldots, \alpha_J$ *are complex numbers, as are the coefficients $\beta(j,t)$. The t are nonnegative integers.)*

(B) *Φ and Ψ can be continued analytically into the entire s-plane, except for a finite number of poles. Furthermore, Φ and Ψ are bounded in each "lacunary" vertical strip:*

$$\sigma_1\leqq \mathrm{Re}\, s\leqq\sigma_2, \quad |\mathrm{Im}\, s|\geqq t_0>0,$$

and they satisfy the functional equation,

$$\Psi(k-s)=e^{\pi ik/2}C\Phi(s). \tag{2.6}$$

Remarks. (i) The lemma differs somewhat from Bochner's original formulation in [1], which deals with generalized Dirichlet series, rather than the ordinary Dirichlet series we have here. Furthermore, Bochner's period functions $q(z)$ in the modular relation (2.4) are "residual functions" – in his terminology – rather than the sums (2.5). (See also [3].) However, the sums (2.5) are residual in Bochner's sense; moreover, a residual function occurring as a period function in a modular relation (2.4) necessarily has the form (2.5).

(ii) The discrete group relevant to Hamburger's theorem is Γ_ϑ, the subgroup of the modular group $\Gamma(1)=\mathrm{SL}(2,Z)$ generated by $z\to z+2$, $z\to -1/z$, since the inverse Mellin transform of $\pi^{-s}\Gamma(s)\zeta(2s)$ is $\vartheta(z)=\vartheta_3(0\,|\,z)$, a modular form of weight $1/2$ on Γ_ϑ. Consequently, in the proofs of Theorems 1 and 2 we apply Lemma 3 with $\lambda_1=\lambda_2=2$, $k=1/2$ and $F=G$ (thus $\Phi=\Psi$, as well). With $\lambda_1=\lambda_2=2$ and $F=G$, the modular relation (2.4) states that F is a "modular integral" of weight k on Γ_ϑ, with "period function" $q(z)$. (See §III for the definitions.)

Lemma 3 omits any mention of the relationship between the period function $q(z)$ and the Dirichlet series in statement (B). In fact, $q(z)$ determines the exact locations and orders of the poles of Φ and Ψ, and vice versa. We make this precise, for the relevant special case $F=G$, $\Phi=\Psi$, in

Lemma 4 *Suppose* $F=G$, $\Phi=\Psi$, *in Lemma* 3. *Then the term* $\beta z^{\alpha}(\log z)^t$, $\beta\neq 0$, $t\in Z$, $t\geqq 0$, *occurring in* (2.5) *gives rise to poles of* $\Phi(s)$ *of order* $t+1$ *at the points* $s=\alpha+k$ *and* $s=-\alpha$. *The only possible further singularities of* Φ *are simple poles at* $s=0$ *and* $s=k$.

III Modular integrals

1 Period cocyles

The theory we introduce in this section is valid for the class of finitely generated discrete groups Γ (acting on $\mathscr{H}$, say), either of finite or infinite hyperbolic area. Since we require the theory here only for $\Gamma=\Gamma_\vartheta$, we shall restrict ourselves to that case throughout.

Let k be a real number and v a multiplier system for the weight k and the group Γ_ϑ. (For a discussion of multiplier systems see [10, pp. 12–13; 15, pp. 267–268], or [12, p. 608]. Let $\mathscr{P}$ denote the collection of all functions φ holomorphic in $\mathscr{H}$, satisfying the growth condition

$$|\varphi(z)|\leqq K(|z|^{\alpha}+y^{-\beta}),\quad y=\operatorname{Im} z>0, \tag{3.1}$$

for some constants $K, \alpha, \beta>0$.

To introduce the notion of "modular integral" we make some preliminary observations. Suppose f is a function holomorphic in $\mathscr{H}$; define the functions q_M, $M\in\Gamma_\vartheta$, as follows:

$$\bar{v}(M)(cz+d)^{-k}f(Mz)=f(z)+q_M(z),\quad M=\begin{pmatrix}* & *\\ c & d\end{pmatrix}\in\Gamma_\vartheta. \tag{3.2}$$

The q_M are called the **period functions** of f, relative to (Γ_ϑ, k, v). Introducing the customary **stroke operator**

$$(f|_k^v M)(z)=\bar{v}(M)(cz+d)^{-k}f(Mz),\quad M=\begin{pmatrix}* & *\\ c & d\end{pmatrix}\in\Gamma_\vartheta, \tag{3.3}$$

we can rewrite the relations (3.2) as

$$f|_k^v M=f+q_M,\quad \text{for all } M\in\Gamma_\vartheta. \tag{3.4}$$

From (3.4) the **cocyle condition** for $\{q_M\,|\,M\in\Gamma_\vartheta\}$ follows directly:

$$q_{MN}=q_M|N+q_N,\quad \text{for } M, N\in\Gamma_\vartheta. \tag{3.5}$$

(We use the abbreviation $f|_k^v M=f|M$.) A collection of functions $\{q_M\,|\,M\in\Gamma_\vartheta\}$ satisfying (3.5) is called a **cocycle relative to** (Γ_ϑ, k, v).

As in [12], we restrict all cocyles to the space $\mathscr{P}$; this applies, of course, to the q_M occurring in (3.4). In [12, pp. 627–630] I have shown: given $\varphi\in\mathscr{P}$ and

a parabolic linear fractional transformation Q (that is, trace $Q=\pm 2$), then there exists $\rho\in\mathscr{P}$ such that

$$\rho|Q-\rho=\varphi. \tag{3.6}$$

Consequently, we can apply a standard argument using (3.4) to derive "Fourier expansions" for f at the parabolic cusps, $i\infty$ and -1, of Γ_ϑ. Note that the stabilizer of $i\infty$ in Γ_ϑ is generated by S^2 and that of -1 by $S^{-2}T$, where $S=\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$, $T=\left(\begin{smallmatrix}0 & -1\\ 1 & 0\end{smallmatrix}\right)$ are the generators of $\Gamma(1)$. (S^2, T generate Γ_ϑ.)

The expansion at $i\infty$ has the form

$$f(z)=\rho_0(z)+\sum_{n=-\infty}^{\infty} a_n e^{\pi i(n+\varkappa)z},\quad \operatorname{Im} z>0, \tag{3.7a}$$

with complex a_n. Here ρ_0 is a function in $\mathscr{P}$ determined by (3.6), with $\varphi=q_{S^2}$ and $Q=S^2$, and $\varkappa$ is a parameter attached to Γ_ϑ and the multiplier system $v: v(S^2)=e^{2\pi i\varkappa}$, $0\leqq\varkappa<1$. The expansion at -1 has the form

$$f(z)=\rho_1(z)+(z+1)^{-k}\sum_{n=-\infty}^{\infty} b_n \exp\{-2\pi i(n+\mu)/(z+1)\}, \tag{3.7b}$$

with complex b_n. In (3.7b), ρ_1 is an element of $\mathscr{P}$ arising from (3.6), this time with $\varphi=q_{S^{-2}T}$ and $Q=S^2$, and μ is determined from v by: $v(S^{-2}T)=e^{2\pi i\mu}$, $0\leqq\mu<1$. (See [10, pp. 17–23 and 75–81].)

Remarks. (i) We make the simple, but important, observation that an exponential series of the form (3.7.a) lies in the space $\mathscr{P}$ if and only if the coefficients a_n satisfy

(a) the polynomial growth condition (2.2)

and

(b) $a_n=0$ for $n<0$.

(The same holds true for exponential series of the form (3.7.b), if we substitute b_n for a_n.) This enters into the proof of our theorems at the conclusion of the argument (§IV).

(ii) The observation in Remark (i) has the unpleasant consequence that the expansions in (3.7) are not unique: the coefficients a_n, b_n with $n\geqq 0$ can be changed, with a corresponding change in $\rho_0(z)$, $\rho_1(z)$. The coefficients a_n, b_n with $n<0$ are uniquely determined by f, however. Of course, if $q_{S^2}=0$ we choose $\rho_0=0$ in (3.7.a); similarly, if $q_{S^{-2}T}=0$ we choose $\rho_1=0$ in (3.7.b).

2 *Definition of "modular integral"*

Assume that $\{q_M|M\in\Gamma_\vartheta\}$ is a cocycle relative to (Γ_ϑ, k, v), with $Q_M\in\mathscr{P}$. Suppose f is a function holomorphic in $\mathscr{H}$ satisfying (3.4), and such that the

expansions (3.7) are left-finite; that is, suppose there exist $n_0, n_1 \in Z$ such that summation condition is $\sum_{n=n_0}^{\infty}$ in (3.7.a) and $\sum_{n=n_1}^{\infty}$ in (3.7.b). (Though the expansions (3.7) are not unique, the condition of left-finiteness is well-defined. In particular, as noted above, the "principal parts" in (3.7.a, b) are unique.) Then we call f **a modular integral with respect to Γ_ϑ, of weight k and multiplier system v, with period functions – or period cycle –** $\{q_M \mid M \in \Gamma_\vartheta\}$. If we can choose $n_0, n_1 \geqq 0$ in (3.7), we call f an **entire** modular integral. If $q_M = 0$ for all $M \in \Gamma_\vartheta$, f is called a **modular form** (or **entire modular form**) rather than a modular integral (entire modular integral).

3 Construction of modular integrals

As we have observed, a holomorphic function engenders a cocycle $\{q_M\}$ of period functions (but not necessarily in $\mathscr{P}$), with respect to the given weight and multiplier system. Eichler [4, 5] introduced the "generalized Poincaré series" to deal with the converse: to construct a modular integral f satisfying (3.4) with a given cocycle $\{q_M\}$ relative to (Γ_ϑ, k, v).

To describe Eichler's construction, begin with a cocycle $\{q_M\}$ for (Γ_ϑ, k, v), subject to the further restriction $q_{S^2} = 0$; let $m \in 2Z^+$. The **generalized Poincaré series** $\Psi(\{q_M\}, m; z) = \Psi(z)$ is defined by

$$\Psi(z) = \sum_V q_V(z)(cz+d)^{-m}, \tag{3.8}$$

where the summation is on all $V = \left(\begin{smallmatrix} * & * \\ c & d \end{smallmatrix}\right) \in \Gamma_\vartheta$, with distinct lower rows c, d. The condition $q_{S^2} = 0$ guarantees that the individual terms of the series (3.8) depend only upon the lower row c, d of $V \in \Gamma_\vartheta$. Eichler shows that for m sufficiently large, the series (3.8) converges absolutely for z in $\mathscr{H}$, and uniformly on compact subsets of $\mathscr{H}$. Eichler's proof assumes that the q_V are polynomials, but a slight elaboration of his method establishes the convergence for the more general case $q_V \in \mathscr{P}$ $(V \in \Gamma)$ [12, pp. 615–619]. In either case (q_V polynomials or $q_V \in \mathscr{P}$) it is required that the group in question be discrete and of finite hyperbolic area (hence finitely generated). This, of course, includes Γ_ϑ.

As a consequence of absolute convergence $\Psi(z)$ has the transformation property

$$(\Psi \mid M)(z) = (\gamma z + \delta)^m \Psi(z) - (\gamma z + \delta)^m E_m(z) q_M(z), \tag{3.9}$$

$$\text{for } M = \begin{pmatrix} * & * \\ \gamma & \delta \end{pmatrix} \in \Gamma_\vartheta,$$

where $E_{mn}(z)$ is the familiar Eisenstein series of weight m,

$$E_m(z) = \sum_V (cz+d)^{-m}\ , \tag{3.10}$$

with summation condition as in (3.8). Since we have assumed that m is an even integer, $E_m(z)$ does not vanish identically. Thus we may form the quotient $F(z) = -\Psi(z)/E_m(z)$, which, by (3.9) and well-known transformation properties of $E_m(z)$ (a modular form of weight m on Γ_ϑ), has the formal behavior of a modular integral relative to (Γ_ϑ, k, v), with the preassigned cocycle $\{q_M\}$ of period functions:

$$F|M = F + q_M, \quad M \in \Gamma_\vartheta. \tag{3.11}$$

However, F may not be a modular integral in the sense of §III.2, since the zeros of $E_m(z)$ in $\mathscr{H}$ are likely poles of F. There is the further issue of appropriate behavior of F at the parabolic cusps of Γ_ϑ; that is to say, we must show that the expansions (3.7) of F (the existence of which derive from (3.11)) are left-finite. As it turns out, the left-finiteness is entailed by the following three facts:

(i) Ψ is in $\mathscr{P}$ [12, Proposition 7, p. 619];
(ii) $E_m(z) \to 2$, as $z \to i\infty$ within a fundamental region for Γ_ϑ;
(iii) $E_m(z)$ has a zero of finite order (in the appropriate exponential variable) at the cusp -1.

The proofs of Theorems 1 and 2 are largely procedures for modifying F to obtain entire modular integrals relative to (Γ_ϑ, k, v).

IV Sketch of the Proofs

1 Brief overview

In order to apply Lemma 3 we make the preliminary observation that, in the presence of the functional equation (condition 2 of the Introduction), the growth restriction contained in condition 1 ($P(s)\varphi(s)$ has finite genus) is equivalent to the apparently stronger boundedness condition of part (B), Lemma 3. This follows from a standard application of Stirling's formula and the Phragmén–Lindelöf Principle.

By Lemmas 3 and 4 (§II) the proofs reduce to establishing the existence of modular integrals f on Γ_ϑ, with period cocycles $\{q_M\}$ such that (i) $q_{S^2} = 0$; (ii) $q_T = q$ is a log-polynomial sum (2.5) with suitable α_j and coefficients $\beta(j, t)$; (iii) the expansion (3.7.a) of f has the form

$$f(z) = \sum_{n=0}^{\infty} a_n e^{\pi i n z}, \tag{4.1}$$

with coefficients a_n that satisfy the growth condition (2.2). In §III.1 we observed that f has the representation (4.1) with a_n satisfying the condition (2.2) if and only if f is in $\mathscr{P}$. Furthermore, by [11] an entire modular integral on Γ_ϑ,

with period functions in $\mathscr{P}$, is itself in $\mathscr{P}$. Ultimately, then, the proofs entail proving the existence of entire modular integrals on Γ_9, with $q_{S^2}=0$ and q_T a suitably prescribed log-polynomial sum.

2 Proof of Theorem 2

The expansion (4.1) suggests an entire modular integral on Γ_9 with respect to a multiplier system v such that $v(S^2)=1$ (i.e., the parameter $\varkappa=0$). Since $T^2=I$ is the only relation in Γ_9, we may – and do – impose the condition $v(S^2)=1$. By standard properties of multiplier systems, $T^2=I$ implies that $v(T)=\pm e^{-\pi ik/2}$. According to Lemma 3, we must choose $v(T)=e^{-\pi ik/2}$ in the proof of our theorems.

Fix a log-polynomial sum $q(z)$ satisfying

$$q|T+q=0. \tag{4.2}$$

(Let $q=\varphi|T-\varphi$, with φ any log-polynomial sum.) Then define a cocycle $\{q_M|M\in\Gamma_9\}$ relative to (Γ_9, k, v) by putting $q_{S^2}=0$, $q_T=q$ and letting the cocycle condition (3.5) generate q_M for arbitrary $M\in\Gamma_9$, after writing M as a word in S^2 and T. Note that since Γ_9 has the unique relation $T^2=I$, condition (4.2) is necessary and sufficient for this procedure to yield well-defined q_M.

As in §III.3, we employ the cocycle $\{q_M\}$ to form a function F satisfying (3.11). This is possible since $q_M\in\mathscr{P}$, for all M in Γ_9, and $q_{S^2}=0$. If F were an entire modular integral on Γ_9, then Theorem 2 would follow from Lemmas 3 and 4, inasmuch as the log-polynomial sum $q_T=q$ is arbitrary, except for the condition (4.2). However, (4.2) is equivalent, on the Dirichlet series side of the correspondence, to the condition $A(k-s)=A(s)$ satisfied by the pole terms of $R(s)$ in Theorem 2.

To complete the proof of Theorem 2, we consider once again the function

$$F(z)=-\Psi(\{a_M\}, m; z)/E_m(z).$$

Clearly, the only possible singularities of F in a fixed fundamental region $\mathscr{R}$ for Γ_9 are poles at the finitely many zeros of E_m in $\mathscr{R}$. Since $k\geqq 2$, we may invoke the "Mittag-Leffler theorem" for modular forms: there exists a modular form f of weight k and multiplier system v on Γ_9 such that $\hat{F}(z)=F(z)-f(z)$ is an entire modular integral relative to (Γ_9, k, v) [12, 15, 16, 17]. Since f is a modular form, $\hat{F}$ has the same cocycle of periods $\{q_M\}$ as does F. Theorem 2 follows.

3 Proof of Theorem 1

In this case the weight $k=1/2<2$, so that the Mittag–Leffler theorem does not apply. Nevertheless, the proof is similar in principle. Once again we choose

v such that $v(S^2)=1$, $v(T)=e^{-\pi ik/2}=e^{-\pi i/4}$, we generate a cocycle $\{q_M\}$ relative to $(\Gamma_\vartheta, \frac{1}{2}, v)$ from a given log-polynomial sum q and we use $\{q_M\}$ to form the function F. This time, instead of subtracting a suitable modular form from F to obtain an entire modular integral with the same period cocycle, we take linear combinations of different F's – formed from different cocycles $\{q_M\}$ – to obtain entire modular integrals, with modified period cocycles.

In the statement of Theorem 1, the poles of the Dirichlet series are restricted to the vertical strip $1/2-\sigma_0 \leqq \mathrm{Re}\, s \leqq \sigma_0$. By Lemma 4 this is equivalent to restricting the exponents α_j in the corresponding log-polynomial sum $q=q_T$ to the strip $-\sigma_0 \leqq \mathrm{Re}\, s \leqq \sigma_0 - 1/2$. From the manner in which the cocycle $\{q_M\}$ is generated from q, and the calculations of [12, §§II.2–3, §III.1], it follows that for log-polynomial sums (2.5) with α_j's, restricted to $-\sigma_0 \leqq \mathrm{Re}\, s \leqq \sigma_0 - 1/2$ we can choose the parameter m in $F=-\Psi/E_m$ to be any even integer $>2\sigma_0+7/2$.

Consequently, fixing σ_0 and any collection of log-polynomial sums satisfying (4.3), with α_j's in $-\sigma_0 \leqq \mathrm{Re}\, s \leqq \sigma_0 - 1/2$, we can choose the same m for all of the F's we form from the cocycles $\{q_M\}$ arising from these log-polynomial sums. Since each such F has precisely the same E_m in the denominator, the poles of all of the F's in the fundamental region $\mathscr{R}$ of Γ_ϑ are restricted to the zeros of this E_m – hence to a fixed finite set of points – with total order at most $m/4$. (This, since $m/4$ is the total order of the zeros of E_m in $\mathscr{R}$.)

The integer $N(\sigma_0)$ in the statement of Theorem 1, then, is the number of linear conditions we must impose upon the collection of F's we derive, in order to eliminate all poles in $\mathscr{R}$ – thus to obtain a linear combination which is an **entire** modular integral. Since the only requirement upon m is that it be an even integer greater than $2\sigma_0+7/2$, we may always choose $m \leqq [2\sigma_0+7/2]+2$. The total order of the zeros of E_m in $\mathscr{R}$ is, then,

$$m/4 \leqq [2\sigma_0+7/2]/4+1/2,$$

so we choose $N(\sigma_0)$ to be any integer greater than $[2\sigma_0+7/2]/4+1/2$. $N(\sigma_0)=[\sigma_0/2+3/8]+2$ suffices. (An argument involving the order of the zero of E_m at the elliptic fixed point i would improve the choice of $N(\sigma_0)$ slightly.)

But linearly independent q_T correspond to linearly independent F, and linearly independent F to linearly independent Φ. Thus, Theorem 1 follows from Lemmas 3 and 4.

V Extensions

The proof of Theorem 1 can be carried over to any weight $k<2$ and both corresponding multiplier systems in weight k. For $k \geqq 2$, of course, the stronger Theorem 2 applies.

There are surely analogues of Theorems 1 and 2 for the Hecke groups $G(\lambda_p)$, generated by $S(\lambda_p)=\left(\begin{smallmatrix}1 & \lambda_p\\ 0 & 1\end{smallmatrix}\right)$, $T=\left(\begin{smallmatrix}1 & -1\\ 1 & 0\end{smallmatrix}\right)$, where $\lambda_p=2\cos\pi/p$, $p\in Z$, $p\geqq 3$. However, in contrast to Γ_ϑ, $G(\lambda_p)$ has the two relations $T^2=(S(\lambda_p)T)^p=I$, and

this creates difficulties in generating the required cocycles $\{q_M\}$. Indeed, in order to formulate suitable analogues of Theorems 1 and 2, we must determine which log-polynomial sums (3.5) satisfy the two relations,

$$q|T+q=0, \quad q|\{[S(\lambda_p)T]^{p-1}+[S(\lambda_p)T]^{p-2}+\cdots+I\}=0.$$

Like Γ_3, the Hecke groups $G(\lambda)$, with $\lambda>2$, have only the single relation $T^2=I$; thus the above difficulty with $\{q_M\}$ does not arise in extending our theorems to this case. The problem here is that for $\lambda>2$, a fundamental region for $G(\lambda)$ has infinite hyperbolic area, so that Eichler's proof of the convergence of the generalized Poincaré series does not apply to this case. In work just completed, Mark Sheingorn and I have overcome this difficulty, proving Theorem 2, with k an arbitrary real weight, for the case $\lambda>2$ [13].

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